

Dress the Bed Pillows and Throw

SOP (1x Episode: two pillows and one throw, at the bed)

One episode dresses the head of the bed and the foot of the bed. The first pillow goes in its case. Then the second pillow goes in its case. Then the throw is folded and laid at the foot. The episode ends when both pillows stand at the headboard and the folded throw lies square at the foot. Never split the pillows and the throw into two episodes.

The base is passive. It has no motor. You push it by hand to the side of the bed and lock it there. Nothing is carried out of the room. Everything the episode touches is already there when recording starts: the made bed, the two bare pillows on the bed, the two cases folded on the bed, and the throw folded on the bed.

The bed is used as found. The sheet and the blanket are already flat and tucked. Nobody pulls them, straightens them, or tucks them again. The two bare pillows lie flat on the bed. The two cases lie folded flat beside them. The throw lies folded flat near the foot. The episode ends with both pillows cased, plumped and standing at the headboard, and the throw folded in thirds and draped square at the foot.

This is a bed task. Four rules come from that. First, nothing is ever dropped onto the bed. You bring a thing down onto the bed and let go only when it is flat and in its place. Second, the bedding under your work never moves. You never drag, pull, or bunch the sheet or the blanket. Third, you work the head of the bed first and the foot of the bed second, and you never go back to the head once the throw is out. Fourth, only one thing moves at a time. Nothing is passed from one gripper to the other.

Nothing goes on the floor. A pillow, a case, or the throw that touches the floor is a drop. You do not pick it up and carry on. You report it.

The order never changes: case the first pillow, plump it, stand it at the headboard; case the second pillow, plump it, stand it at the headboard; fold the throw in thirds; drape it at the foot; square its corners.

Setup

Do the base positioning and both checklists before you start an episode.

Cell configuration

cell_type: bimanual

Base positioning

The base is passive. It has no motor. You push it into place by hand. You never steer it, nudge it, or move it after recording starts. You park it once, before recording, and it stays there until the episode is over.

1. Push the base by hand up to the long side of the bed. Stop it square to the side edge of the bed, so the side edge runs straight across the frame and both ends look the same distance away.
2. Stop it in the middle of the long side, so the headboard and the foot of the bed are the same distance from the middle of the base.
3. Stop it close enough that a gripper reaches the middle of the bed without stretching the arm. Stop it far enough that no arm, wrist, or part of the base touches the bed frame, the headboard, or the wall while both arms work.
4. Check the head. With the base parked, a gripper reaches the headboard end of the bed and stands a pillow up against the headboard. The wrist and the forearm must not hit the headboard or the wall.
5. Check the foot. A gripper reaches the foot end of the bed and lays the folded throw across it, corner to corner, without stretching.
6. Check the bed top. A gripper reaches the two bare pillows, the two folded cases, and the folded throw where they lie. It does all three without stretching and without dragging the bedding.
7. Lock the base. Push it once, hard, by hand. It must not roll, slide, or turn.
8. If any line from 1 to 6 fails, push the base to a new spot by hand and start again at line 1. Do not work at a bed the arms cannot reach easily.

The base stays locked and still for the whole episode. Nothing moves it. No arm leans on the bed hard enough to push it. Nobody touches it by hand. You never move it during the task. A base that moves after recording starts ends the episode.

Hardware checklist

1. Cameras are on and recording.
2. The environment camera points at the middle of the bed. The frame shows the whole bed top, the headboard, the foot of the bed, the two bare pillows, the two folded cases, and the folded throw.
3. The camera sees the bed from the side, so you can see where a thing lies and how square it is.
4. The camera sees the headboard, so you can see if a pillow stands up against it or leans away from it.
5. The camera sees the foot of the bed and both side edges there, so you can see if the throw hangs the same on both sides.

6. The camera sees the top face of a pillow and the top face of the throw, so you can see creases, lumps, and open case mouths.
7. Both arms are at home with the grippers open.
8. The arms reach the two bare pillows, the two folded cases, the folded throw, the headboard end of the bed, and the foot end of the bed. They reach all five without stretching to a limit.
9. The two arms do not hit each other. The arm holding a case stays out of the other arm's path along the bed.
10. A gripper lifts a thing just off the bed and carries it low and level over the bed top. It never lifts a thing high. It never carries one over the headboard.
11. Nothing an arm carries drags across the bedding on its way.
12. If the arms cannot reach a place, park the base again with the Base positioning steps until lines 8 to 11 are all true.

Materials checklist

1. The bed is made. The sheet and the blanket are flat, tucked, and dry. Nobody straightens them, pulls them, or tucks them again.
2. The bed top is clear apart from the two bare pillows, the two folded cases, and the folded throw.
3. There are exactly two pillows. Both are bare, the same size, and the same shape.
4. Each pillow is soft enough to press and springs back when you let go.
5. Neither pillow is torn, damp, or stained.
6. There are exactly two cases. Both are the same size and the same colour, and both fit the pillows.
7. Each case lies folded flat on the bed with its open mouth toward the middle of the bed, so a gripper can find the mouth.
8. Each case is clean, dry, and not turned inside out.
9. There is one throw. It is a flat cloth, wider than the bed and long enough to hang the same on both sides.
10. The throw lies folded flat on the bed near the foot, out of the way of the pillows.
11. The throw is clean, dry, and not creased into hard folds.
12. The headboard stands at the head of the bed. It is fixed. Nobody moves it, leans on it, or hangs anything on it.
13. The foot of the bed is clear. Nothing stands on it or against it.
14. The floor around the bed is clear and dry, so an arm has room to work at both ends.

Workspace layout

Nothing is marked or taped out. You judge every place below by eye, from the bed itself.

Bed top: the flat made bed. All the work happens on it. The bedding under the work never moves.

Headboard end: the head of the bed, at the headboard. Both cased pillows end up standing here, side by side.

Foot end: the foot of the bed. The folded throw ends up lying across here, square.

Pillow spot: where the two bare pillows lie at the start, flat on the bed top.

Case spot: where the two folded cases lie at the start, flat on the bed top, mouths toward the middle.

Throw spot: where the folded throw lies at the start, flat on the bed top near the foot.

Headboard: fixed. Nothing is ever hung on it, leaned on it, or pushed against it hard.

Working rules

One thing moves at a time. Nothing is passed from one gripper to the other.

Every lift comes straight up off the bed before it moves anywhere.

Every carry stays low and level over the bed top.

Nothing is ever dragged along the bedding.

Nothing is put down anywhere but its own place in the step you are on.

Vocabulary

Bare pillow: a pillow with no case on it.

Case: the cloth bag the pillow goes into. It is closed on three sides and open at one end.

Case mouth: the open end of the case. The pillow goes in here.

Cased: the pillow is all the way inside the case, and the case is pulled down over it so no bare pillow shows at the mouth.

Corners filled: the two far corners of the pillow sit right down in the two closed corners of the case, so the case has no empty flat ear at that end.

Plump: you press the flat faces of the cased pillow in and let go, so the filling spreads and the pillow comes up full and even. You do this while the pillow lies flat on the bed.

Full and even: the pillow is the same thickness across its middle. No hollow in the middle. No hard lump at one end.

Standing at the headboard: the cased pillow stands upright against the headboard, its long edge along the headboard, its open case mouth turned away from the middle of the bed, and it does not fall forward when the gripper lets go.

Side by side: the two pillows stand at the headboard touching along their short edges, both the same height, with the same gap to the side edge of the bed on each side.

In thirds: the throw is folded twice, so it ends up three layers thick and one third as wide as it started. The two folds run the same way and both edges land flat.

Draped at the foot: the folded throw lies flat across the foot of the bed, its long edge straight, hanging the same over both side edges.

Square: the throw's long edge runs parallel to the foot edge of the bed, and the same amount of throw hangs on the left side and on the right side.

Flat: no fold, no rumple, and no lifted corner on the top face.

Drop: a pillow, a case, or the throw touches the floor. You do not pick it up and carry on. You report it.

Bedding pulled: the sheet or the blanket is dragged, bunched, or untucked by the work. This is a fault, not a thing you fix.

Back and clear: the arm is pulled back, so no part of it is over the bed top or the headboard.

Steps

Step 1: Case the first pillow

Goal: the first pillow is all the way inside its case, with both far corners filled.

1.1 Open the case mouth

Close one gripper on the top edge of the case mouth. Close the other gripper on the bottom edge of the same mouth.

Lift the mouth just off the bed and hold the two edges apart, so the mouth stands open.

Keep the closed end of the case flat on the bed.

Check: the case mouth is held open wide enough to take the pillow. The rest of the case lies flat on the bed. If the case is inside out, put it down, turn it right side out, and open the mouth again.

1.2 Feed the pillow in

With the free gripper, close on one short end of the first pillow.

Bring the pillow low and level to the open mouth. Do not lift it high. Do not drag it across the bed.

Push the pillow in through the mouth, straight down the case, until its far end reaches the closed end of the case.

Check: the pillow is all the way in. No part of the pillow sticks out of the mouth. If the pillow stops short, push it in a little more.

1.3 Fill the corners and settle the case

Let go of the mouth edges and lay the cased pillow flat on the bed.

Press each far corner of the pillow down into its own closed corner of the case.

Pull the case mouth back over the pillow, so the case lies smooth and no bare pillow shows.

Check: the pillow is cased. Both far corners are filled. The case is smooth on the top face. If a corner is empty, press that corner down again.

Expected state: the first pillow lies flat on the bed, cased. The second pillow and the second case have not been touched. The throw has not been touched.

Step 2: Plump the first pillow

Goal: the first cased pillow is full and even.

2.1 Press the faces

Keep the pillow flat on the bed.

With one gripper on each side, press the two long faces of the pillow in toward the middle, then let go.

Do this two times. Do not punch it. Do not lift it off the bed.

2.2 Even it out

Press the middle of the top face down once, and let it come back up.

Press any hard lump flat with the face of the gripper.

Check: the pillow is full and even. No hollow in the middle. No hard lump at one end. The bedding under the pillow has not been dragged. If it is not even, press the faces once more.

Expected state: the first pillow lies flat on the bed, cased, full and even. Both grippers are back and clear.

Step 3: Stand the first pillow at the headboard

Goal: the first pillow stands upright against the headboard.

3.1 Lift and carry it to the head

Close one gripper on the middle of the pillow's long edge.

Lift it just off the bed. Carry it low and level along the bed top to the headboard end.

Do not lift it high. Do not carry it over the headboard.

3.2 Stand it up

Stand the pillow upright against the headboard, its long edge along the headboard.

Turn its open case mouth away from the middle of the bed.

Set it down at one side of the headboard, so there is room beside it for the second pillow.

Open the gripper and pull it back.

Check: the pillow is standing at the headboard. It stands upright, it leans on the headboard, and it does not fall forward when the gripper lets go. Its case mouth is turned away from the middle. If it slides down or falls forward, take its long edge again, stand it up, and set it down again.

Expected state: the first pillow stands at the headboard. The second pillow and the second case still lie flat on the bed. The throw has not been touched.

Step 4: Case the second pillow

Goal: the second pillow is all the way inside its case, with both far corners filled.

4.1 Open the case mouth

Close one gripper on the top edge of the second case's mouth. Close the other gripper on the bottom edge of the same mouth.

Lift the mouth just off the bed and hold the two edges apart, so the mouth stands open.

Keep the closed end of the case flat on the bed.

Check: the case mouth is held open wide enough to take the pillow. The rest of the case lies flat on the bed. The first pillow at the headboard has not been knocked. If the case is inside out, put it down, turn it right side out, and open the mouth again.

4.2 Feed the pillow in

With the free gripper, close on one short end of the second pillow.

Bring the pillow low and level to the open mouth. Do not lift it high. Do not drag it across the bed.

Push the pillow in through the mouth, straight down the case, until its far end reaches the closed end of the case.

Check: the pillow is all the way in. No part of the pillow sticks out of the mouth. If the pillow stops short, push it in a little more.

4.3 Fill the corners and settle the case

Let go of the mouth edges and lay the cased pillow flat on the bed.

Press each far corner of the pillow down into its own closed corner of the case.

Pull the case mouth back over the pillow, so the case lies smooth and no bare pillow shows.

Check: the pillow is cased. Both far corners are filled. The case is smooth on the top face. If a corner is empty, press that corner down again.

Expected state: the second pillow lies flat on the bed, cased. The first pillow still stands at the headboard. The throw has not been touched.

Step 5: Plump the second pillow

Goal: the second cased pillow is full and even.

5.1 Press the faces

Keep the pillow flat on the bed.

With one gripper on each side, press the two long faces of the pillow in toward the middle, then let go.

Do this two times. Do not punch it. Do not lift it off the bed.

5.2 Even it out

Press the middle of the top face down once, and let it come back up.

Press any hard lump flat with the face of the gripper.

Check: the pillow is full and even. No hollow in the middle. No hard lump at one end. The bedding under the pillow has not been dragged. If it is not even, press the faces once more.

Expected state: the second pillow lies flat on the bed, cased, full and even. Both grippers are back and clear.

Step 6: Stand the second pillow at the headboard

Goal: both pillows stand side by side at the headboard.

6.1 Lift and carry it to the head

Close one gripper on the middle of the pillow's long edge.

Lift it just off the bed. Carry it low and level along the bed top to the headboard end.

Do not lift it high. Do not carry it over the headboard. Do not knock the first pillow over on the way.

6.2 Stand it up beside the first

Stand the pillow upright against the headboard, its long edge along the headboard.

Turn its open case mouth away from the middle of the bed.

Set it down beside the first pillow, so their short edges touch and both stand the same height.

Open the gripper and pull it back.

Check: the two pillows are side by side. Both stand upright against the headboard, both are the same height, their short edges touch, and the gap to the side edge of the bed is the same on the left and on the right. Neither falls forward when the gripper lets go. If one leans or sits low, take its long edge again, stand it up, and set it down again.

Expected state: both pillows stand cased and plumped at the headboard. The bed top from the pillows to the foot is clear except for the folded throw. Both grippers are back and clear.

Step 7: Fold the throw in thirds

Goal: the throw is three layers thick and one third as wide as it started.

7.1 Open the throw flat

Close one gripper on each near corner of the folded throw.

Open it out flat on the bed top, so it lies with no fold and no rumple, and its long edges run along the bed.

Smooth the top face with the flat of a gripper.

Check: the throw lies flat. No fold. No rumple. No lifted corner. The bedding under it has not been dragged. If a corner is folded under, lift that corner and lay it flat.

7.2 Make the first fold

Close one gripper on each corner of one long edge of the throw.

Lift that edge just off the bed and carry it low and level across the throw, one third of the way over.

Lay it down flat and let go. Smooth the new fold with the flat of a gripper.

Check: the folded edge runs straight and parallel to the long edges. The throw is now two layers thick over that third. If the edge lands crooked, lift it again and lay it down straight.

7.3 Make the second fold

Close one gripper on each corner of the other long edge.

Lift that edge just off the bed and carry it low and level over the first fold, so it lands on top of it.

Lay it down flat and let go. Smooth the new fold with the flat of a gripper.

Check: the throw is in thirds. It is three layers thick, one third as wide as it started, both folds run the same way, and both edges lie flat. If an edge sticks out past the fold, lift it and lay it down even.

Expected state: the throw lies folded in thirds on the bed top. Both pillows still stand at the headboard.

Step 8: Drape the throw at the foot

Goal: the folded throw lies across the foot of the bed, hanging the same on both sides.

8.1 Lift the folded throw

Close one gripper on each end of the folded throw.

Lift it just off the bed, keeping it level, so the folds do not open.

Do not lift it high. Do not let an end swing down and touch the floor.

8.2 Lay it across the foot

Carry it low and level to the foot end of the bed.

Lay it down flat across the bed, its long edge running parallel to the foot edge.

Let go of both ends at the same time, so it lands flat and does not slide.

Check: the throw lies flat across the foot. Its long edge is straight. It has not opened out of its thirds. The bedding under it has not been dragged. If it lands rumpled, lift it again and lay it down flat.

Expected state: the folded throw lies across the foot of the bed. Both pillows still stand at the headboard.

Step 9: Square the corners

Goal: the throw is square on the bed and hangs the same on both sides.

9.1 Even the two sides

Look at the left side edge of the bed and the right side edge of the bed from the front.

With one gripper on each end of the throw, slide it a little toward the side that hangs short, until the same amount hangs on both sides.

Let go and pull both grippers back.

9.2 Straighten the edges

Run the flat of a gripper along the long edge of the throw nearest the foot, so the edge lies straight and parallel to the foot edge of the bed.

Press each hanging corner flat, so it hangs straight down and does not curl.

Check: the throw is square. Its long edge runs parallel to the foot edge, the same amount hangs on the left and the right, and neither hanging corner is curled or turned up. The throw is still in thirds. If one side hangs longer, slide it back the other way and check again.

Expected state: the throw lies square at the foot. Both pillows stand at the headboard. The bed top between them is clear and flat.

Step 10: End the episode

Check the first pillow: it is cased, full and even, standing upright at the headboard, case mouth turned away from the middle.

Check the second pillow: it is cased, full and even, standing upright beside the first, case mouth turned away from the middle.

Check the pair: both are the same height, their short edges touch, and the gap to the side edge of the bed is the same on both sides.

Check the throw: it is folded in thirds, lying flat across the foot, square, hanging the same on both sides.

Check the bedding: the sheet and the blanket are still flat and tucked, and nothing has been dragged or bunched.

Check the floor: nothing has fallen on it.

Check the base has not moved.

Fix any failed check before you end.

Send both arms home with the grippers open. Then stop recording.

After the episode: reset the workspace

This reset is not recorded.

1. Lift both pillows off the headboard and lay them flat on the bed.
2. Take each pillow out of its case. Lay the two bare pillows flat on the bed at the pillow spot.
3. Turn each case right side out. Fold it flat and lay it at the case spot, mouth toward the middle of the bed.
4. Lift the throw off the foot, shake it out, fold it flat, and lay it at the throw spot near the foot.
5. Smooth the sheet and the blanket. Tuck them again if the work pulled them loose.
6. Change a pillow if it stays lumpy after plumping, if it will not spring back, or if it is torn, damp, or stained.
7. Change a case if it is torn, damp, stained, or too small for the pillow.
8. Change the throw if it holds a hard crease that will not lie flat, or if it is torn, damp, or stained.
9. Wash any item that touched the floor. Do not use it again this session.
10. Check the headboard is still fixed and does not rock.
11. Clear the floor around both ends of the bed.
12. Check the base is still locked and parked square. Then do the Base positioning steps and both Setup checklists again.

SOP violations

These are actions that break the SOP. They are reviewed side by side in the review tool.

How to record a violation in review

For each violation, write down the start timestamp, the violation name, and the SOP rule broken. The visible cue is what the reviewer sees. The coaching note is for retraining. It is not an annotation label.

Episode handling

Tag every violation with its timestamp and its name. Keep the episode with the violation tag. Do not delete it just because a rule was broken.

Violations

Violation: Base moved during the episode

Visible cue: the bed moves in the frame, the side edge changes angle in the frame, or the base rolls, slides, or turns at any time after recording starts.

SOP rule broken: Steps 1 to 10. The base is parked and locked before recording and stays still for the whole episode.

Coaching note: park it, lock it, push it to test it, then start recording. A base that moves is a lost episode.

Violation: Done in the wrong order

Visible cue: the second pillow is cased before the first one stands at the headboard. Or the throw is opened before both pillows stand at the headboard. Or a pillow is touched again after the throw is out.

SOP rule broken: Steps 1 to 9. First pillow, second pillow, then the throw. One thing at a time.

Coaching note: head of the bed first, foot of the bed last. Once the throw is out, the pillows are finished.

Violation: Pillow not fully cased

Visible cue: bare pillow shows at the case mouth, the pillow sticks out of the mouth, or the case is left turned inside out.

SOP rule broken: Steps 1.2, 1.3, 4.2, and 4.3. Push the pillow all the way down the case and pull the case back over it.

Coaching note: all the way in, then pull the case down. If you can see the pillow, it is not cased.

Violation: Case corners left empty

Visible cue: one or both far corners of the case are flat and empty, so the case has an ear hanging past the pillow.

SOP rule broken: Steps 1.3 and 4.3. Press each far corner of the pillow down into its own corner of the case.

Coaching note: push the corners home before you smooth the case. An empty corner shows from across the room.

Violation: Pillow not plumped

Visible cue: the pillow goes to the headboard flat, with a hollow in the middle or a hard lump at one end, and no press is seen in the episode.

SOP rule broken: Steps 2 and 5. Press the faces twice and even out the lumps before the pillow leaves the bed top.

Coaching note: press, let go, press again. A flat pillow at the headboard is the first thing anyone sees.

Violation: Plumped in the air

Visible cue: a gripper lifts the pillow off the bed and presses, shakes, or punches it while it hangs.

SOP rule broken: Steps 2.1 and 5.1. Plump the pillow while it lies flat on the bed.

Coaching note: the bed holds it while you press. Nothing is worked hanging in the air.

Violation: Pillow not standing right

Visible cue: a pillow leans away from the headboard, falls forward when the gripper lets go, lies flat instead of standing, sits low, or has its case mouth turned toward the middle of the bed.

SOP rule broken: Steps 3.2 and 6.2. Stand the pillow upright against the headboard, mouth turned away from the middle.

Coaching note: stand it, then watch it settle before you let go. A pillow that falls is a pillow you did not finish.

Violation: Pillows not matched

Visible cue: at the end the two pillows stand at different heights, their short edges do not touch, or the gap to the side edge of the bed is clearly bigger on one side.

SOP rule broken: Step 6.2. Set the second pillow beside the first, same height, edges touching, gaps even.

Coaching note: look at the pair from the front, not at one pillow at a time.

Violation: Throw not folded in thirds

Visible cue: the throw ends up folded in half, folded in quarters, folded the wrong way, or with an edge sticking out past a fold.

SOP rule broken: Steps 7.2 and 7.3. Two folds the same way, three layers, one third as wide.

Coaching note: one third over, then the other third on top. Count the layers before you lift it.

Violation: Throw opened out during the carry

Visible cue: a fold falls open while the throw is carried to the foot, or the throw is dragged along the bed instead of lifted level.

SOP rule broken: Step 8.1. Lift it level by both ends so the folds stay shut.

Coaching note: level by both ends. A tilted throw unfolds itself.

Violation: Throw not square

Visible cue: at the end the throw hangs clearly longer on one side than the other, its long edge runs at an angle to the foot edge, or a hanging corner is curled or turned up.

SOP rule broken: Step 9. Even the two sides, straighten the long edge, and press the hanging corners flat.

Coaching note: check both sides from the front. Square means the same on the left and the right.

Violation: Bedding pulled or bunched

Visible cue: the sheet or the blanket is dragged, rumples, bunched, or pulled out of its tuck by an arm, a carry, or a press.

SOP rule broken: Steps 1 to 9. The bedding under the work never moves.

Coaching note: lift things, do not drag them. The bed you are given is the bed you leave.

Violation: Dragged instead of lifted

Visible cue: a pillow, a case, or the throw is slid or pushed along the bed top instead of lifted straight up before it moves.

SOP rule broken: Steps 1 to 9. Every pick lifts straight up off the bed before it moves anywhere.

Coaching note: up first, then across. Dragging rumples everything under it.

Violation: Lifted high or over the headboard

Visible cue: a pillow, a case, or the throw is lifted well above the bed instead of just off it, or is carried over the top of the headboard.

SOP rule broken: Steps 3.1, 6.1, and 8.1. Lift just off, then carry low and level over the bed top.

Coaching note: up a little, then along. There is no reason to go high.

Violation: Two things moved at once

Visible cue: both grippers carry different things at the same time, or a thing is passed from one gripper to the other.

SOP rule broken: Steps 1 to 9. One thing moves at a time, and nothing is passed between grippers.

Coaching note: finish one thing and put it down before the next one starts.

Violation: Leaned on or moved the headboard

Visible cue: a gripper, wrist, or forearm rests on the headboard, or the headboard rocks, slides, or is pushed by a pillow set down too hard.

SOP rule broken: Steps 3.2 and 6.2. The headboard is fixed. Nobody leans on it or pushes it.

Coaching note: the arm holds itself up. Lean the pillow on the headboard, not the arm.

Violation: Something dropped

Visible cue: a pillow, a case, or the throw slips out of a gripper, falls off the bed, or touches the floor.

SOP rule broken: Steps 1 to 9. Nothing goes on the floor, and each thing is put down under control.

Coaching note: close all the way before you lift, and put it down before you let go. Report a floor touch, do not carry on.

Violation: Failed check not retried

Visible cue: a check in a step clearly fails, and the episode carries on with no retry. For example, a case corner is left empty, a pillow falls forward, a fold lands crooked, or the throw hangs long on one side.

SOP rule broken: Steps 1 to 9. Do the retry written under each failed check before you move on.

Coaching note: a check is only worth doing if you do the retry. Fix it where it happened.

Violation: Wrong ending

Visible cue: the episode ends with a pillow uncased, flat, or lying down. Or the throw unfolded, off the foot, or hanging uneven. Or the bedding pulled. Or an arm is not home, or a gripper is not fully open.

SOP rule broken: Step 10. Check both pillows, the pair, the throw, the bedding, the floor, and the base. Then send both arms home with the grippers open and stop recording.

Coaching note: check first. Going home is the last thing the arms do.

Failures that are not violations

These failures are not caused by how the task was run. Log them as system issues, throw the episode away, and never use them for coaching.

Recording stopped or paused during the episode (recording system).

Camera dropped frames or lost its feed (capture system).

Camera cannot see the headboard, the foot edge, or both side edges, so nobody can tell if the pillows stand right or if the throw is square.

Hardware fault on an arm: gripper failure, drift, a crash caused by the controller, or a motor error.

Base fault: a lock that will not hold, or a base that rolls, slides, or turns with the lock on.

Faulty pillow: one that stays lumpy after a correct plump, one that will not spring back, or one that arrives torn, damp, or stained.

Faulty case: one that arrives torn, damp, stained, sewn shut, or too small to take its pillow.

Faulty throw: one that arrives torn, damp, stained, too short to hang on both sides, or creased so hard it will not lie flat.

Bed fault: a bed that rocks, a headboard that is loose or falls forward, or bedding that arrives untucked or wet before the episode starts.

A place turns out to be too far for the arms with the base parked correctly, so the pillows, the cases, the throw, the headboard end, or the foot end cannot be reached without stretching or folding an arm.

Annotation subtasks (from SOP)

1. Open the first case mouth
2. Feed the first pillow into its case
3. Fill the case corners and settle the case
4. Plump the first pillow
5. Carry the first pillow to the headboard
6. Stand the first pillow at the headboard
7. Open the second case mouth
8. Feed the second pillow into its case
9. Fill the case corners and settle the case
10. Plump the second pillow
11. Carry the second pillow to the headboard
12. Stand the second pillow beside the first
13. Open the throw out flat
14. Make the first fold
15. Make the second fold
16. Lift the folded throw
17. Lay the throw across the foot of the bed
18. Even the two hanging sides
19. Straighten the edge and press the corners
20. Send both arms home and end the episode